2.42 Parametric Equations: Differentiation

Worked Solutions 1–23

1)

$t = x + 1$ so $y = 2(x+1)^2$.

$$\frac{dy}{dx} = 4(x+1) = 4x + 4$$

2 a)

$\frac{dx}{dt} = 1$, $\frac{dy}{dt} = -4t$ so $\frac{dy}{dx} = -4t$.

2 b)

$\frac{dx}{dt} = 2t + 3$, $\frac{dy}{dt} = 3t^2 + 2t$ so $\frac{dy}{dx} = \frac{3t^2+2t}{2t+3}$.

2 c)

$\frac{dx}{dt} = 2e^{2t}$, $\frac{dy}{dt} = 2e^t$ so $\frac{dy}{dx} = \frac{2e^t}{2e^{2t}} = \frac{e^t}{e^{2t}} = \frac{1}{e^t}$.

2 d)

$\frac{dx}{dt} = 8e^{2t}$, $\frac{dy}{dt} = -3e^{-t}$ so $\frac{dy}{dx} = \frac{-3e^{-t}}{8e^{2t}} = -\frac{3}{8e^{3t}}$.

2 e)

$\frac{dx}{dt} = 1 + \sin t$, $\frac{dy}{dt} = 1 - \cos t$ so $\frac{dy}{dx} = \frac{1-\cos t}{1+\sin t}$.

3)

$\frac{dx}{dt} = -\frac{2t}{(1+t^2)^2}$, $\frac{dy}{dt} = 2t$.

$$\text{so } \frac{dy}{dx} = -\frac{2t(1+t^2)^2}{2t} = -(1+t^2)^2$$

4)

$\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 2t$, so $\frac{dy}{dx} = 2t$. When $t = 2$, $\frac{dy}{dx} = 4$, $x = 3$, $y = 4$.

$$y - 4 = 4(x-3) \text{ so } y = 4x - 8$$

5)

$\frac{dx}{dt} = 3t^2$, $\frac{dy}{dt} = 2$, so $\frac{dy}{dx} = \frac{2}{3t^2}$. When $t = 3$, $\frac{dy}{dx} = \frac{2}{27}$, $x = 27$, $y = 6$.

$$y - 6 = -\frac{27}{2}(x-27) \text{ so } y = -\frac{27}{2}x + \frac{741}{2}$$

6)

$\frac{dx}{dt} = e^t$, $\frac{dy}{dt} = -\frac{1}{t^2}$, so $\frac{dy}{dx} = -\frac{1}{t^2 e^t}$. When $t = 1$, $\frac{dy}{dx} = -\frac{1}{e}$, $x = e$, $y = 1$.

$$y - 1 = e(x-e) \text{ so } y = ex - e^2 + 1$$

7 a)

$\frac{dx}{dt} = 4t^3 + 1$, $\frac{dy}{dt} = 10 - 10t$, so $\frac{dy}{dx} = \frac{10-10t}{4t^3+1}$. Stat point when $\frac{dy}{dx} = 0$, so $10 - 10t = 0$. Therefore $t = 1$. Giving $(2, 5)$.

7 b)

$\frac{dx}{dt} = 6$, $\frac{dy}{dt} = 3t^2 - 10t + 3$, so $\frac{dy}{dx} = \frac{3t^2-10t+3}{6}$. Stat point when $\frac{dy}{dx} = 0$, so $3t^2 - 10t + 3 = 0$. Therefore $t = \frac{1}{3}$ or $t = 3$. Giving $(2, \frac{13}{27})$ or $(18, -9)$.

7 c)

$\frac{dx}{dt} = e^{2t}$, $\frac{dy}{dt} = 12t - 4t^2 - 5$, so $\frac{dy}{dx} = \frac{12t-4t^2-5}{e^{2t}}$. Stat point when $\frac{dy}{dx} = 0$, so $-4t^2 + 12t - 5 = 0$. Therefore $t = \frac{1}{2}$ or $t = \frac{5}{2}$. Giving $(e, -\frac{7}{6})$ or $(e^5, \frac{25}{6})$.

8 a)

To maximise y, maximise $(t + 2)^2$.

$$t = 1 \Rightarrow (2, 9)$$

8 b)

To minimise y, minimise $(t + 2)^2$, $t = -1 \Rightarrow (4, 1)$.

8 c)

$\frac{dx}{dt} = -3t^2$, $\frac{dy}{dt} = 2(t + 2)$ so $\frac{dy}{dx} = -\frac{2(t+2)}{3t^2}$.

9 a)

$\frac{dx}{dt} = -4t^{-2}$, $\frac{dy}{dt} = 3$, so $\frac{dy}{dx} = -\frac{3}{4}t^2$. At $(2, 5)$, $t = 2$ so $\left.\frac{dy}{dx}\right|_{t=2} = -\frac{3}{4} \times 4 = -3$.

$$y - 5 = -3(x - 2) \text{ so } y = -3x + 11$$

9 b)

Area is $\frac{1}{2} \times \frac{11}{3} \times 11 = \frac{121}{6}$ units^2.

10)

$\frac{dx}{dt} = t^{-2}$, $\frac{dy}{dt} = 4t - 2$, so $\frac{dy}{dx} = (4t - 2)t^2 = 4t^3 - 2t^2$. At $x = -2$, $t = \frac{1}{3}$.

$$t = \tfrac{1}{3} \text{ so } \left.\frac{dy}{dx}\right|_{t=\frac{1}{3}} = -\frac{2}{27}$$

At $t = \frac{1}{3}$, $y = -\frac{4}{9}$.

$$y + \frac{4}{9} = \frac{27}{2}(x + 2) \text{ so } 18y - 243x = 478$$

11 a)

When $t = 0.5$, $y = 6.3$ metres.

11 b)

When $t = 3$, $y = \frac{7}{15} = 0.467$ metres (3 SF).

11 c)

$x = \frac{35}{8} = 4.375$ metres.

11 d)

Lowest point of the slide is when $\frac{dy}{dx} = 0$.

$$\frac{dx}{dt} = t, \quad \frac{dy}{dt} = -5t^{-2} + 1 \text{ so } \frac{dy}{dx} = \frac{-5t^{-2} + 1}{t}$$

$$\Rightarrow -\frac{5}{t^2} + 1 = 0 \Rightarrow t = \sqrt{5}$$

$$y = \frac{5}{\sqrt{5}} + \sqrt{5} - \frac{21}{5} = 0.272 \text{ metres}$$

$6.3 - 0.272 = 6.03$ metres (3 SF).

12 a)

$t = x - a$ so $y = (x - a)^2 + 3$.

12 b)

$\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 2t$ so $\frac{dy}{dx} = 2t \Rightarrow \left.\frac{dy}{dx}\right|_{t=1} = 2$. When $t = 1$, $x = a + 1$, $y = 4$.

$$y - 4 = -\frac{1}{2}(x - (a + 1)) \text{ so } y = -\frac{1}{2}x + \frac{a + 9}{2}$$

13 a)

$\frac{dx}{d\theta} = \sqrt{3}\cos\theta$ and $\frac{dy}{d\theta} = 4\cos 2\theta$.

$$\text{so } \frac{dy}{dx} = \frac{4\cos 2\theta}{\sqrt{3}\cos\theta}$$

13 b)

When $\theta = \frac{\pi}{3}$, $x = \frac{3}{2}$, $y = \sqrt{3}$.

$$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{3}} = \frac{-2}{\frac{\sqrt{3}}{2}} = -\frac{4}{3}\sqrt{3}$$

$$y - \sqrt{3} = \frac{3}{4\sqrt{3}}(x - \tfrac{3}{2}) \Rightarrow 4\sqrt{3}y - 12 = 3x - \tfrac{9}{2}$$

$$\Rightarrow 8\sqrt{3}y - 6x = 15$$

13 c)

When $x = 0$, $y = \frac{15}{8\sqrt{3}}$. When $y = 0$, $x = -\frac{15}{6}$.

$$\text{Area is } \frac{1}{2} \times \frac{15}{8\sqrt{3}} \times \frac{15}{6} = \frac{75}{32\sqrt{3}} = \frac{25\sqrt{3}}{32}$$

14 a)

$\frac{dx}{dt} = -2\ln 3 \times 3^{-t}$, $\frac{dy}{dt} = 4\ln 3 \times 3^{t}$.

$$\text{so } \frac{dy}{dx} = -\frac{4\ln 3 \times 3^{t}}{2\ln 3 \times 3^{-t}} = -\frac{2 \times 3^{t}}{3^{-t}} = -2 \times 3^{2t}$$

14 b)

When $x = 0$,

$$2 \times 3^{-t} + 4 = 0 \Rightarrow 3^{-t} = -2$$

which has no solutions, therefore the curve does not cross the y-axis.

14 c)

$\frac{x-4}{2} = 3^{-t}$ so $\frac{2}{x-4} = 3^{t}$.

$$y = 4\left(\frac{2}{x-4}\right) - 6 \qquad y = \frac{8}{x-4} - 6$$

14 d)

As $t \to \infty$, $x \to 4$.

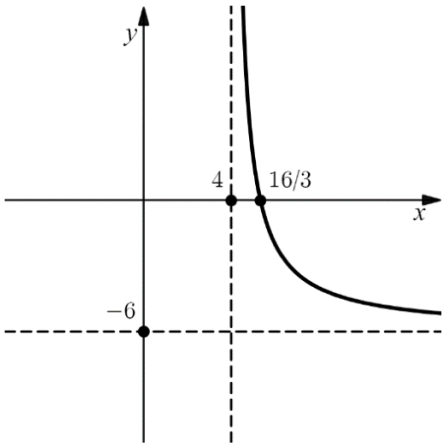
14 e)

As $t \to \infty$, $y \to -6$.

14 f)

When $y = 0$, $0 + 6x - 0 = 32$. Therefore $x = \frac{16}{3}$ so at $(\frac{16}{3}, 0)$.

14 g)



15 a)

$\frac{dx}{d\theta} = 4\cos 4\theta$ and $y = (\cos\theta)^{-2}$.

$$\frac{dy}{d\theta} = -2(\cos\theta)^{-3} \times -\sin\theta = \frac{2\sin\theta}{\cos^3\theta}$$

$$\text{so } \frac{dy}{dx} = \frac{2\sin\theta}{4\cos 4\theta \cos^3\theta} = \frac{\sin\theta}{2\cos 4\theta \cos^3\theta}$$

15 b)

When $y = 2$, $\frac{1}{\cos^2\theta} = 2 \Rightarrow \cos^2\theta = \frac{1}{2}$.

$$\Rightarrow \cos\theta = \pm\frac{1}{\sqrt{2}} \text{ for } 0 \leq \theta \leq \frac{\pi}{2}, \text{ then } \theta = \frac{\pi}{4}$$

$$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{4}} = \frac{\frac{\sqrt{2}}{2}}{2 \times -1 \times (\frac{\sqrt{2}}{2})^3} = -1$$

16 a)

$\frac{dx}{dt} = 15t^2$, $\frac{dy}{dt} = 4t^3$ so $\frac{dy}{dx} = \frac{4t^3}{15t^2} = \frac{4}{15}t$.

$$\begin{aligned} y - t^4 &= \frac{4}{15}t(x - 5t^3) \\ 15y - 15t^4 &= 4tx - 20t^4 \\ 15y - 4tx + 5t^4 &= 0 \text{ as req} \end{aligned}$$

16 b)

When $x = 40$, $t = 2$ so $15y - 8x + 80 = 0$. When $t = 2$, $y = 16$. Tangent line meets the x-axis at $x = 10$. Area is $\frac{1}{2} \times (40 - 10) \times 16 = 240$ units^2.

17 a)

$\frac{dx}{d\theta} = \sec^2\theta$, $\frac{dy}{d\theta} = \sec\theta\tan\theta$.

$$\text{so } \frac{dy}{dx} = \frac{\sec\theta\tan\theta}{\sec^2\theta} = \frac{\tan\theta}{\sec\theta} = \sin\theta$$

$$\begin{aligned} y - (\sec a) &= -\frac{1}{\sin a}(x - (\tan a + 1)) \\ y - (\sec a) &= -\frac{1}{\sin a}x + \frac{1 + \tan a}{\sin a} \\ y &= -\frac{1}{\sin a}x + \frac{1 + \tan a}{\sin a} + \sec a \\ y &= -\frac{1}{\sin a}x + \operatorname{cosec} a + 2\sec a \end{aligned}$$

17 b)

Minimum is when $\frac{dy}{dx} = \sin a = 0 \Rightarrow a = 0$. When $\theta = 0$, $x = \tan 0 + 1 = 1$, $y = \sec 0 = 1$. So minimum at $(1, 1)$.

18 a)

$\frac{dx}{dt} = 4t^3$, $\frac{dy}{dt} = 6t^2 + 8t^3$.

$$\text{so } \frac{dy}{dx} = \frac{6t^2 + 8t^3}{4t^3} = \frac{3 + 4t}{4t}$$

18 b)

Stat pts when $\frac{dy}{dx} = 0$ so $3 + 4t = 0$, so when $t = -\frac{3}{4}$ which is at $(\frac{81}{256}, -\frac{27}{128})$.

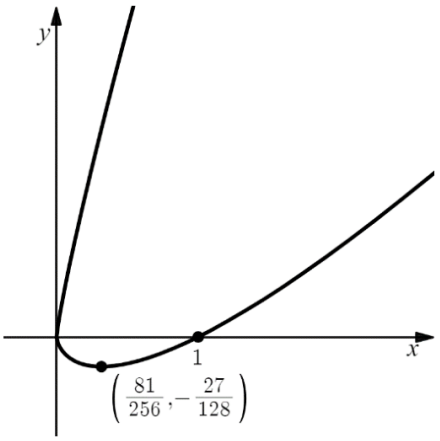
18 c)

When $x = 0$, $t = 0$ so $y = 0$, therefore $(0, 0)$.

18 d)

When $y = 0$, $t = 0$ or $t = -1$ so $x = 0$ or $x = 1$. Therefore $(0, 0)$ and $(1, 0)$.

18 e)



19 a)

When $t = 0$, $p = \frac{1}{2} - \frac{1}{2}\cos 0 = 0$. When $t = 0$, $q = \frac{1}{2} - \frac{1}{4}\sin 0 = \frac{1}{2}$. 0.5 metres.

19 b)

$$\begin{aligned} p - q &= \tfrac{1}{2} - \tfrac{1}{2}\cos t - (\tfrac{1}{2} - \tfrac{1}{4}\sin t) \\ &= -\tfrac{1}{2}\cos t + \tfrac{1}{4}\sin t \end{aligned}$$

$$R\sin(t - \alpha) = R\sin t\cos\alpha - R\cos t\sin\alpha$$

$R\cos\alpha = \frac{1}{4}$ and $R\sin\alpha = \frac{1}{2}$.

$$\therefore R = \frac{\sqrt{5}}{4} \text{ and } \alpha = \tan^{-1}\left(\frac{1/2}{1/4}\right) = 1.11 \text{ (3 SF)}$$

So $\frac{\sqrt{5}}{4}\sin(t - 1.11)$.

19 c)

$\sin(t - 1.11) = 1$ for max, so maximum distance is $\frac{\sqrt{5}}{4}$. So 0.56 metres.

20 a)

$\frac{dx}{d\theta} = -4\sin\theta$ and $\frac{dy}{d\theta} = 5\cos\theta$.

$$\text{so } \frac{dy}{dx} = -\frac{5\cos\theta}{4\sin\theta}$$

20 b)

$y - 5\sin\theta = -\frac{5\cos\theta}{4\sin\theta}(x - 4\cos\theta)$.

20 c)

$$\begin{aligned} 12 - 5\sin\theta &= -\frac{5\cos\theta}{4\sin\theta}(4 - 4\cos\theta) \\ 48\sin\theta - 20\sin^2\theta &= -20\cos\theta + 20\cos^2\theta \\ 48\sin\theta + 20\cos\theta &= 20\sin^2\theta + 20\cos^2\theta \\ 48\sin\theta + 20\cos\theta &= 20 \\ 12\sin\theta + 5\cos\theta &= 5 \end{aligned}$$

20 d)

$$\begin{aligned} R\sin(\theta + \alpha) &= R\sin\theta\cos\alpha + R\cos\theta\sin\alpha \\ R\sin(\theta + \alpha) &= 12\sin\theta + 5\cos\theta \end{aligned}$$

So $\cos\alpha = 12$ and $\sin\alpha = 5$. $R = \sqrt{144 + 25} = 13$. $\tan\alpha = \frac{5}{12} \Rightarrow \alpha = 0.39479\ldots$.

$$\begin{aligned} 13\sin(\theta + 0.39479\ldots) &= 12\sin\theta + 5\cos\theta \\ 13\sin(\theta + 0.39479\ldots) &= 5 \\ \sin(\theta + 0.39479\ldots) &= \tfrac{5}{13} \end{aligned}$$

$\theta = 0$. $\theta = 2.352010414\ldots$ so $\theta = 2.35$.

21 a)

$\frac{dx}{d\theta} = -a\sin\theta$ and $\frac{dy}{d\theta} = 4a\cos\theta$.

$$\text{so } \frac{dy}{dx} = -\frac{4a\cos\theta}{a\sin\theta} = -\frac{4\cos\theta}{\sin\theta}$$

$$\begin{aligned} y - 4a\sin\theta &= -\frac{4\cos\theta}{\sin\theta}(x - a\cos\theta) \\ y\sin\theta - 4a\sin^2\theta &= -4x\cos\theta + 4a\cos^2\theta \\ y\sin\theta + 4x\cos\theta &= 4a\sin^2\theta + 4a\cos^2\theta \\ y\sin\theta + 4x\cos\theta &= 4a(\sin^2\theta + \cos^2\theta) \\ y\sin\theta + 4x\cos\theta &= 4a \text{ as req} \end{aligned}$$

21 b)

$$\begin{aligned} 3a\sin\theta + 4a\cos\theta &= 4a \\ 3\sin\theta + 4\cos\theta &= 4 \\ R\sin(\theta + \alpha) &= R\sin\theta\cos\alpha + R\cos\theta\sin\alpha \\ R\sin(\theta + \alpha) &= 3\sin\theta + 4\cos\theta \end{aligned}$$

So $\cos\alpha = 3$ and $\sin\alpha = 4$. $R = \sqrt{9 + 16} = 5$ and $\tan\alpha = \frac{4}{3} \Rightarrow \alpha = 0.927295218\ldots$.

$$\begin{aligned} 5\sin(\theta + 0.927\ldots) &= 3\sin\theta + 4\cos\theta \\ 5\sin(\theta + 0.927\ldots) &= 4 \end{aligned}$$

21 c)

$\sin(\theta + 0.927\ldots) = \frac{4}{5}$. $\theta = 0$. $\theta = 1.287002218\ldots$ so $\theta = 1.29$.

22 a)

$t = 0$ gives $(0, 1)$ and $t = \frac{\pi}{4}$ gives $(1, 0)$. $y - 0 = -(x - 1)$ so $y = -x + 1$.

22 b)

$\frac{dx}{dt} = \sec^2 t$ and $\frac{dy}{dt} = -2\sin 2t$.

$$\begin{aligned} \text{so } \frac{dy}{dx} &= \frac{-2\sin 2t}{\sec^2 t} = -2\sin 2t\cos^2 t \\ &= -4\sin t\cos t\cos^2 t = -4\sin t\cos^3 t \end{aligned}$$

22 c)

$f(x) = 4\sin x\cos^3 x - 1$. $f'(x) = -12\sin^2 x\cos^2 x + 4\cos^4 x$.

$$x_{n+1} = x_n - \frac{4\sin x_n\cos^3 x_n - 1}{-12\sin^2 x_n\cos^2 x_n + 4\cos^4 x_n}$$

$$\begin{aligned} x_1 &= 0.25 \\ x_2 &= 0.2852070084\ldots \\ x_3 &= 0.2874040157\ldots \\ x_3 &= 0.2874131486\ldots \\ x_3 &= 0.2874131488\ldots \end{aligned}$$

So the solution is $x = 0.2874$ to 4 decimal places.

22 d)

$y + x = k$ has gradient -1. So solve $-4\sin t\cos^3 t = -1$. $4\sin t\cos^3 t = 1$ has solution $t = 0.2874$. So at $(0.296, 0.839)$. $y - 0.839 = -(x - 0.296) \Rightarrow y + x = 1.135$. $1 \leq k \leq 1.135$.

23 a)

$\frac{dx}{dt} = \frac{1}{8}\cos t$ and $\frac{dy}{dt} = -t\sin t + \cos t$.

$$\frac{dy}{dx} = \frac{\cos t - t\sin t}{\frac{1}{8}\cos t} = 8\left(\frac{\cos t - t\sin t}{\cos t}\right)$$

23 b)

Stat pt when $\frac{dy}{dx} = 0$. $\cos t - t\sin t = 0$ so $1 - \frac{1}{2}t^2 - t^2 = 0$. $t^2 = \frac{2}{3}$ and given $t \geq 0$, so $t = \sqrt{\frac{2}{3}}$. Coordinate of stat point is approximately at

$$\left(\tfrac{1}{8}\sin\sqrt{\tfrac{2}{3}}, \sqrt{\tfrac{2}{3}}\cos\sqrt{\tfrac{2}{3}}\right) = (0.0911, 0.559)$$

(3 SF). NOTE: by using the tan small angle approximation, a different approximation will be obtained.